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EXPERIMENT NO. : 1

DATE: 13/08/2026

1. AIM

To design and study a lithography mask pattern for Local Oxidation of Silicon, demonstrating selective oxidation for device isolation

2. TOOL

Semulator3D

3. THEORY

Local Oxidation of Silicon (LOCOS) is a semiconductor fabrication technique used to create electrically isolated regions between adjacent devices. In this process, selected areas of a silicon wafer are oxidized to form a thick layer of silicon dioxide, while other regions are protected from oxidation. This selective oxidation is achieved using a suitable masking layer, commonly silicon nitride, which prevents oxygen from reaching the protected silicon surface. A lithography mask pattern defines the regions where oxidation is required. During the fabrication process, the mask protects the active device regions while exposing the isolation regions. Thermal oxidation is then performed, causing silicon dioxide to grow in the exposed regions. The resulting thick field oxide electrically separates neighboring active areas and reduces unwanted current flow between devices. LOCOS is an important isolation technique used in traditional integrated circuit fabrication. The geometry and dimensions of the mask pattern directly determine the location of the isolation regions and active areas. The process also demonstrates the importance of accurate lithographic pattern definition in semiconductor manufacturing. Using Semulator3D, the lithography mask pattern for LOCOS can be designed and studied to visualize the selective oxidation regions. The simulation provides an understanding of how a mask controls oxidation and how the resulting oxide regions contribute to device isolation.

4. DESIGN PARAMETERS

Table 1: DESIGN PARAMETERS

Number	Lock	Action	Name	Material(s)	Dopant(s)	Depth	Mask	Mask Polarity	Wafer	Wafer Side	Lateral Ratio	Parameter Name	Comment
1		Wafer Setup	Wafer Setup	/Silicon/Si_Xtal		400			wafer1				
2		Conformal...	pad oxide depo...	/Oxides/SiO2.T...		15			wafer1	Top	1.0		
3		Conformal...	sin hard mask d...	/Nitrides/Si3N4...		100			wafer1	Top	1.0		
4		Sequence	lithography										
4.1		Deposit	Planarizing...	/Organics/Resist		200			wafer1	Top			
4.2		Expose Mat...	UV exposure	/Organics/Resist		0	LOCOS_Trench	Dark	wafer1	Top			
5		Sequence	etch										
5.1		Enhanced B...	SiN hardmask etch	/Nitrides/, /Org...		100			wafer1	Top	0		
5.2		Enhanced B...	pad oxide etch	/Nitrides/, /Org...		15			wafer1	Top	0.3		
5.3		Remove M...	resist removal	/Organics/Resist					wafer1	Top			
6		Interface G...	thermal oxidation	/Air/, /Silicon/Si_Xtal		200			wafer1	Top			
7		Enhanced B...	SiN hardmask st...	/Nitrides/Si3N4...		105			wafer1	Top	0		
8		Multi Etch (...	Multi Etch	/Oxides/SiO2.T...		10			wafer1	Top			

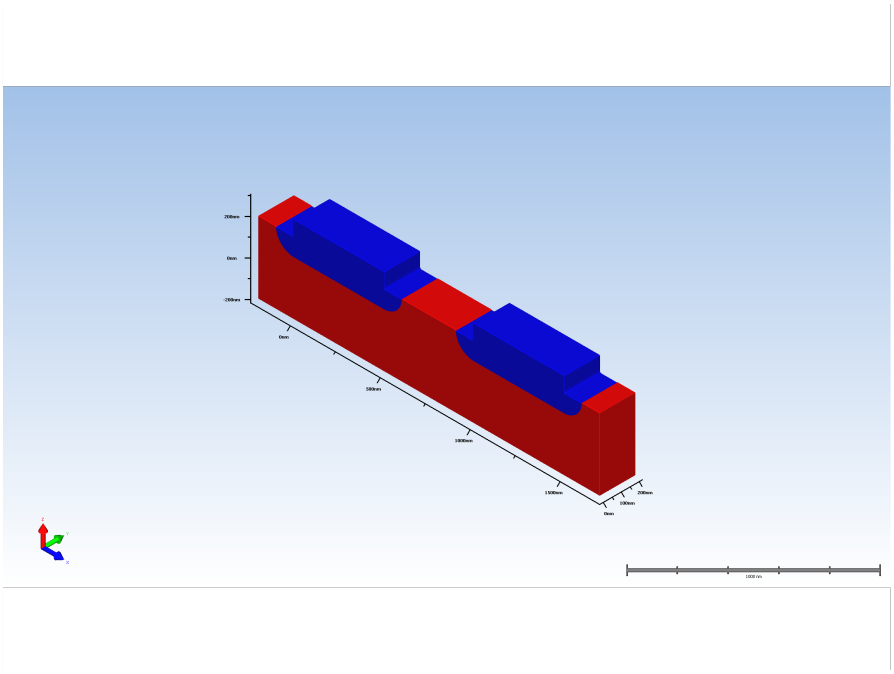


Figure 1: final output

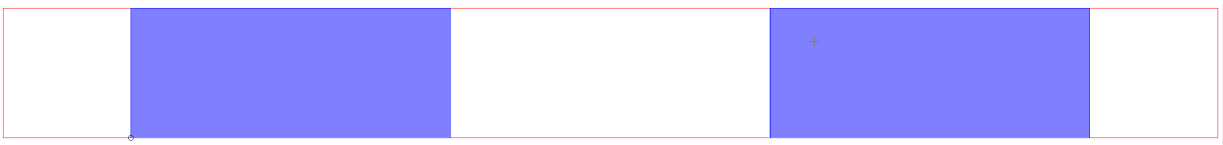


Figure 2: Layout

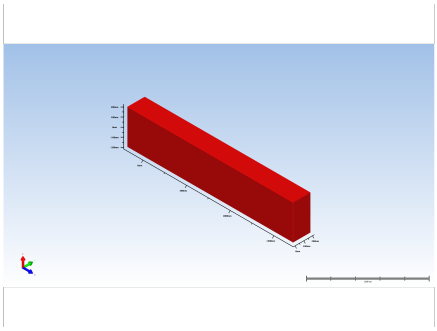


Figure 3: lab3_locus Slide 1

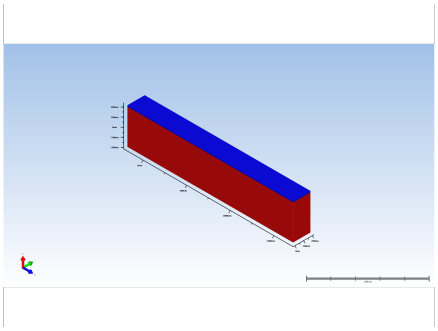


Figure 4: lab3_locus Slide 2

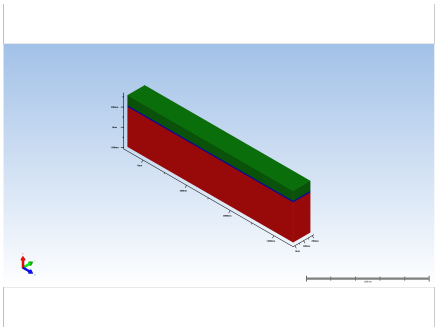


Figure 5: lab3_locus Slide 3

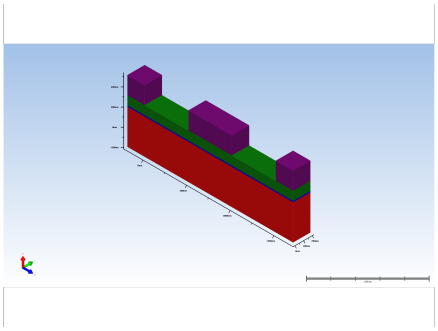


Figure 6: lab3_locus Slide 4

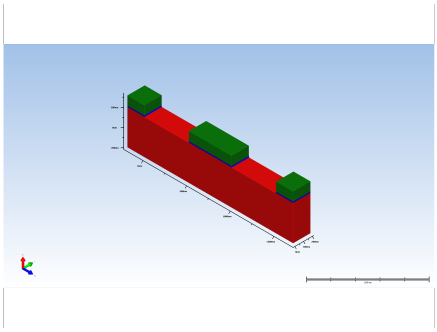


Figure 7: lab3_locus Slide 5

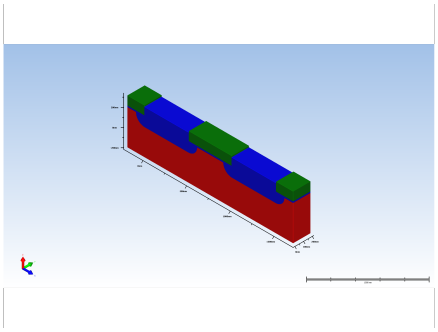


Figure 8: lab3_locus Slide 6

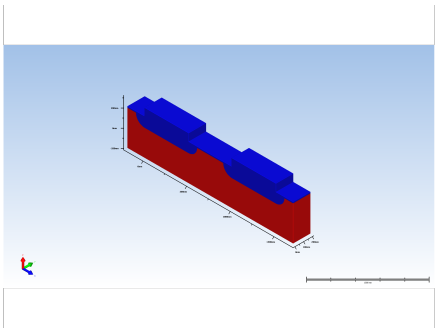


Figure 9: lab3_locus Slide 7

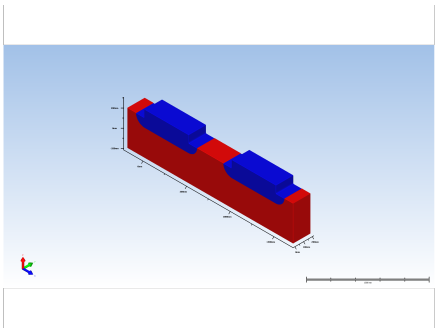


Figure 10: final output

6. OBSERVATION

The lithography mask pattern for Local Oxidation of Silicon was successfully designed and studied using Semulator3D. The pattern clearly defined the regions exposed for oxidation and the protected active regions, demonstrating the principle of selective oxidation used for device isolation.

7. CONCLUSION

The lithography mask pattern for Local Oxidation of Silicon was successfully designed and studied using Semulator3D. The experiment demonstrated selective oxidation by defining protected and exposed regions through the mask pattern. It provided an understanding of how LOCOS forms field-oxide isolation between adjacent active regions during semiconductor fabrication.